Wednesday, Sep 20

In this lecture I will demonstrate how to implement some of the methods we have discussed using the **survey** package for R. To try this yourself you will need to install R. It is highly recommended that you use the RStudio integrated development environment (IDE) when using R. Note that both of these are free and available for a variety of operating systems.

Set-Up

We will be using the **survey** package and a data set included with the **SDaA** package. You can install these packages using the `install.packages` command.

```
install.packages("survey")
install.packages("SDaA")
```

Note that `install.packages` only needs to be done once on a given installation of R. To access the contents of these packages we need to use the `library` command each time we start a R session.

```
library(survey)
library(SDaA)
```

We will use the data set `otters` from the **SDaA** package for a few demonstrations. We can see the first few rows of this data set using `head`.

```
head(otters)
```

```
  section habitat holts
1       1       4     6
2       3       2     0
3       4       1     8
4       8       1     0
5      11       1     0
6      19       2     0
```

These data were collected using stratified random sampling, where the strata are four habitat types (cliff, agricultural, peat, and non-peat). The elements are 5 km by 110 m sections along the coast. The target variable is the number of holts (otter dens).

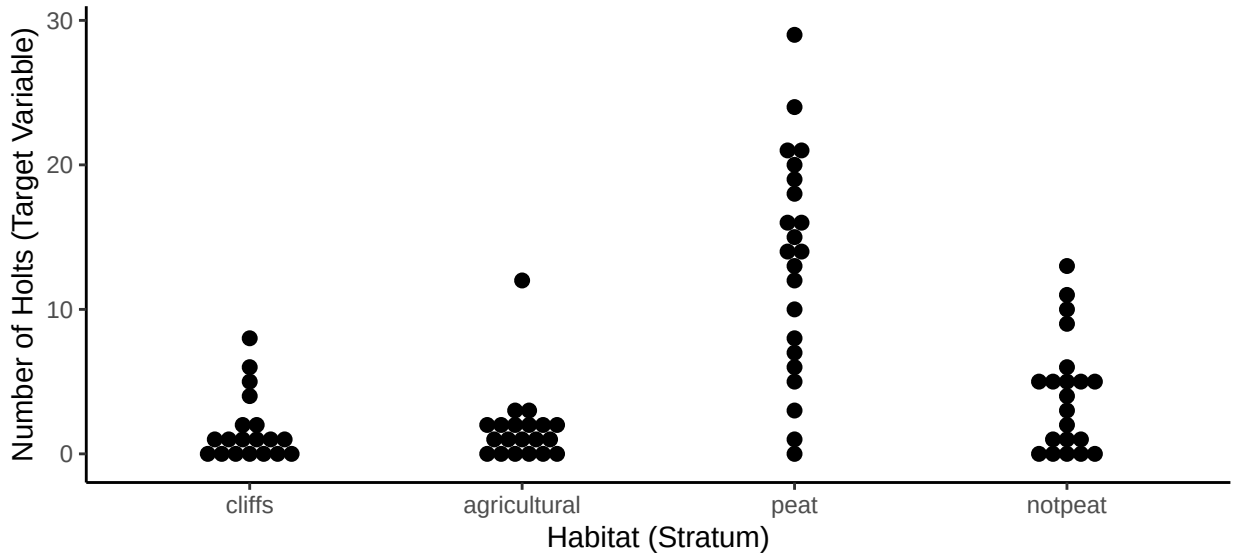
The next step is optional, but useful for keeping track of the strata. Here I change a new variable `habtype` from `habitat` with more clear labels.

```
otters$habtype <- factor(otters$habitat, levels = 1:4,
  labels = c("cliffs","agricultural","peat","notpeat"))
head(otters)
```

```
  section habitat holts      habtype
1       1       4     6      notpeat
2       3       2     0 agricultural
3       4       1     8       cliffs
4       8       1     0       cliffs
5      11       1     0       cliffs
6      19       2     0 agricultural
```

Another optional step, but it might be nice to visualize the data using the **ggplot2** package.

```
library(ggplot2) # install this package with install.packages("ggplot2")
p <- ggplot(otters, aes(x = habtype, y = holts)) + theme_classic() +
  geom_dotplot(binaxis = "y", stackdir = "center") +
  labs(x = "Habitat (Stratum)", y = "Number of Holts (Target Variable)")
plot(p)
```



And yet another optional step would be to compute some basic descriptive statistics using the **dplyr** package.

```
library(dplyr) # install this package with install.packages("dplyr")
otters %>% group_by(habtype) %>% summarize(mean = mean(holts), sd = sd(holts), n = n())
```

```
# A tibble: 4 x 4
  habtype       mean    sd     n
  <fct>        <dbl> <dbl> <int>
1 cliffs        1.74  2.33    19
2 agricultural  1.75  2.61    20
3 peat         13.3   7.67    22
4 notpeat       4.10  3.95    21
```

This computes the sample mean ($\bar{y}_j$), sample standard deviation (s_j), and sample size (n_j) for each sample from each stratum.

Simple Random Sampling

The `otters` data were collected using a *stratified* random sampling design, but for this example we will (incorrectly) assume that the data were collected using *simple* random sampling. Note that data do not know how they were collected. It is the responsibility of the survey researcher to correctly communicate the design to the software.

The first step is to specify the *sampling design*. But before we do that we need to add more information to the data set. Namely we need to specify the *population size* (which is 237 sections). This is done by creating a new variable.

```
otters$N <- 237
head(otters)
```

```
  section habitat holts      habtype   N
1       1       4     6      notpeat 237
```

```
2        3        2     0 agricultural 237
3        4        1     8       cliffs 237
4        8        1     0       cliffs 237
5       11        1     0       cliffs 237
6       19        2     0 agricultural 237
```

Now the design can be specified using the `svydesign` function.

```
ottersrs <- svydesign(id = ~1, data = otters, fpc = ~N)
```

The `id = ~1` is used to indicate a variable that identifies the *sampling unit* for cluster sampling designs (more on that later), but for element sampling we specify it as above. The `data` argument is the data frame containing sample data, and the `fpc` argument is used to indicate the variable that holds the population size(s). The object `ottersrs` now contains information necessary for any inference calculations *assuming a simple random sampling design.*

Recall that an estimator of μ for simple random sampling is the sample mean,

$$\bar{y} = \frac{1}{n} \sum_{i \in \mathcal{S}} y_i.$$

We can compute the estimate from this estimator using the `svymean` function.

```
svymean(~holts, design = ottersrs)
```

```
       mean     SE
holts 5.439 0.6031
```

The `SE` is the (estimated) *standard error*, which is simply the standard deviation of the estimator and thus the square root of the variance of the estimator. The (estimated) variance is computed as

$$\widehat{V}(\bar{y}) = \left(1 - \frac{n}{N}\right) \frac{s^2}{n},$$

where s^2 is the variance of the observations in the sample, defined as

$$s^2 = \frac{1}{n-1} \sum_{i \in \mathcal{S}} (y_i - \bar{y})^2.$$

It is being used here as an estimate of σ^2. We have defined the *bound on the error of estimation* for $\bar{y}$ as (approximately) $B = 2\sqrt{V(\bar{y})}$.[1] So we could obtain the (estimated) bound on the error of estimation by simply doubling the standard error reported by `svymean`. The confidence interval for μ is $\bar{y} \pm B$. We can compute this by passing the result of `svymean` to the `confint` function.

```
confint(svymean(~holts, design = ottersrs))
```

```
         2.5 %   97.5 %
holts 4.256935 6.621114
```

The `confint` function does not necessarily define the bound on the error of estimation as exactly twice the standard error as it tries to use a multiplier to come closer to a 95% confidence level, but in practice it'll usually be close to two. You can also change the confidence level to a different value (e.g., 99%).

```
confint(svymean(~holts, design = ottersrs), level = 0.99)
```

```
         0.5 %   99.5 %
holts 3.885496 6.992553
```

[1] The 'survey' package will try to compute a more accurate multiplier using a t distribution.

Recall that an estimator of τ under simple random sampling is

$$\hat{\tau} = \frac{N}{n} \sum_{i \in \mathcal{S}} y_i = N\bar{y},$$

and the (estimated) variance of $\hat{\tau}$ is

$$\widehat{V}(\hat{\tau}) = N^2 \left(1 - \frac{n}{N}\right) \frac{s^2}{n}.$$

We can use the `svytotal` function for inferences about τ based on this estimator.

```
svytotal(~holts, design = ottersrs)
```

```
      total     SE
holts  1289 142.94
```

```
confint(svytotal(~holts, design = ottersrs))
```

```
         2.5 %   97.5 %
holts 1008.894 1569.204
```

Domain Estimation

To estimate domain means or totals, we can use the `svyby` function.

```
svyby(~holts, by = ~habtype, design = ottersrs, FUN = svymean)
```

```
                  habtype     holts        se
cliffs             cliffs  1.736842 0.4232673
agricultural agricultural  1.750000 0.4634253
peat                 peat 13.272727 1.2994362
notpeat           notpeat  4.095238 0.6841975
```

```
svyby(~holts, by = ~habtype, design = ottersrs, FUN = svytotal)
```

```
                  habtype     holts        se
cliffs             cliffs  95.37805  27.99661
agricultural agricultural 101.15854  31.20485
peat                 peat 843.95122 150.03694
notpeat           notpeat 248.56098  56.33427
```

Recall that we discussed *two* ways to estimate a domain total ($\hat{\tau}_d$) in the context of stratified random sampling, depending on whether or not we know the size of the domain (i.e., N_d). Here `svyby` is using the estimator that does not require knowing the domain size, which is

$$\hat{\tau}_d = \frac{N}{n} n_d \bar{y}_d.$$

If we did know the domain size we could estimate the domain total with the other estimator, which is

$$\hat{\tau}_d = N_d \bar{y}_d.$$

A way to use this estimator is via post-stratification (see below). Confidence intervals for domain means or totals can be obtained by using `confint`.

```
confint(svyby(~holts, by = ~habtype, design = ottersrs, FUN = svytotal))
```

```
                 2.5 %    97.5 %
cliffs        40.50571  150.2504
agricultural  39.99815  162.3189
peat         549.88422 1138.0182
notpeat      138.14783  358.9741
```

Categorical Target Variable

Now suppose instead that target variable was the *habitat type* rather than the number of holts. We can estimate the *proportion* of each habitat type as follows.

```
svymean(~habtype, design = ottersrs)
```

```
                       mean     SE
habtypecliffs       0.23171 0.0379
habtypeagricultural 0.24390 0.0386
habtypepeat         0.26829 0.0398
habtypenotpeat      0.25610 0.0392
```

We can also estimate the *number* of sections of each habitat type as follows.

```
svytotal(~habtype, design = ottersrs)
```

```
                     total     SE
habtypecliffs       54.915 8.9853
habtypeagricultural 57.805 9.1452
habtypepeat         63.585 9.4356
habtypenotpeat      60.695 9.2952
```

Of course this would only be necessary if the researchers did not know how many sections there were of each type.

Stratified Random Sampling

As mentioned earlier, the `otters` data were collected using a *stratified sampling design*, not simple random sampling. Here we will see how to specify this design and make inferences assuming a stratified sampling design.

For stratified random sampling we need to specify the size of *each* stratum.

```
otters$N[otters$habtype == "cliffs"] <- 89
otters$N[otters$habtype == "agricultural"] <- 61
otters$N[otters$habtype == "peat"] <- 40
otters$N[otters$habtype == "notpeat"] <- 47
head(otters)
```

```
  section habitat holts      habtype  N
1       1       4     6      notpeat 47
2       3       2     0 agricultural 61
3       4       1     8       cliffs 89
4       8       1     0       cliffs 89
5      11       1     0       cliffs 89
6      19       2     0 agricultural 61
```

Another way you can do this is with the `case_when` function from the **dplyr** package.

```
otters <- otters %>% mutate(N = case_when(
  habtype == "cliffs" ~ 89,
  habtype == "agricultural" ~ 61,
  habtype == "peat" ~ 40,
  habtype == "notpeat" ~ 47))
head(otters)
```

```
  section habitat holts      habtype  N
1       1       4     6      notpeat 47
2       3       2     0 agricultural 61
```

```
3       4       1     8       cliffs 89
4       8       1     0       cliffs 89
5      11       1     0       cliffs 89
6      19       2     0 agricultural 61
```

Now we can specify the design using `svydesign`.

```
otterstrat <- svydesign(id = ~1, strata = ~habtype, fpc = ~N, data = otters)
```

Inferences can be obtained using the same commands as earlier, although note that the results are not the same because the assumed design is different. Recall that an estimator of μ under stratified random sampling is

$$\hat{\mu} = \frac{N_1}{N}\bar{y}_1 + \frac{N_2}{N}\bar{y}_2 + \cdots + \frac{N_L}{N}\bar{y}_L = \sum_{j=1}^{L} \frac{N_j}{N}\bar{y}_j,$$

which has an estimated variance of

$$\widehat{V}(\hat{\mu}) = \frac{1}{N^2}\sum_{j=1}^{L} N_j^2 \left(1 - \frac{n_j}{N_j}\right)\frac{s_j^2}{n_j}.$$

These formulas are used to obtain the following (recall that the standard error is the square root of the variance).

```
svymean(~holts, design = otterstrat)
```

```
        mean     SE
holts 4.1549 0.3119
```

An estimator of τ under stratified random sampling is

$$\hat{\tau} = N_1\bar{y}_1 + N_2\bar{y}_2 + \cdots + N_L\bar{y}_L = \sum_{i=1}^{L} N_j\bar{y}_j,$$

which has an estimated variance of

$$\widehat{V}(\hat{\tau}) = \sum_{j=1}^{L} N_j^2 \left(1 - \frac{n_j}{N_j}\right)\frac{s_j^2}{n_j}.$$

These formulas are used to obtain the following.

```
svytotal(~holts, design = otterstrat)
```

```
       total     SE
holts 984.71 73.921
```

Confidence intervals can be obtained in the same way as before.

```
confint(svytotal(~holts, design = otterstrat))
```

```
         2.5 %   97.5 %
holts 839.8317 1129.597
```

The optional argument `deff = TRUE` will estimate the *design effect* for the stratified sampling design.

```
svytotal(~holts, design = otterstrat, deff = TRUE)
```

```
        total      SE   DEff
holts 984.714  73.921 0.3572
```

The total sample size is $n = 82$. So the effective sample size of this design is estimated to be $82/0.3572 \approx 230$.

Estimates of domain means and totals can be obtained as follows.

```
svyby(~holts, by = ~habtype, design = otterstrat, FUN = svymean)
```

```
                  habtype     holts        se
cliffs             cliffs  1.736842 0.4739725
agricultural agricultural  1.750000 0.4790589
peat                 peat 13.272727 1.0964954
notpeat           notpeat  4.095238 0.6408521
```

```
svyby(~holts, by = ~habtype, design = otterstrat, FUN = svytotal)
```

```
                  habtype    holts       se
cliffs             cliffs 154.5789 42.18355
agricultural agricultural 106.7500 29.22259
peat                 peat 530.9091 43.85982
notpeat           notpeat 192.4762 30.12005
```

The estimator of μ_d is the same for simple random sampling and stratified random sampling (i.e., $\bar{y}_d$), but the standard errors are not because the design is different. Also here the estimator being used for τ_d is the estimator that uses a *known* N_d, since the domains are the strata and the strata sizes are provided. The formulas for the (estimated) variances of the domain estimators are the same as those for simple random sampling *applied to each sample*.

Further Inferences Concerning Strata

The above shows how to make inferences concerning individual strata parameters. We can also consider (a) how to make inferences about two or more strata *combined* and also (b) how to make inferences about *differences* between strata parameters.

Suppose we wanted to make inferences about the total number of holts in non-agricultural sections (i.e., cliffs, peat, or not peat). There are a couple of ways this can be done. One is to use the `subset` function which communicates that we only want to make inferences about a particular sub-population.

```
notagg <- subset(otterstrat, habtype %in% c("cliffs","peat","notpeat"))
svytotal(~holts, design = notagg)
```

```
       total   SE
holts 877.96 67.9
```

If there was a variable in the original data that identifies the non-agricultural sections we can use that instead.

```
otters <- otters %>%
  mutate(agricultural = ifelse(habtype %in% c("cliffs","peat","notpeat"), "no", "yes"))
head(otters)
```

```
  section habitat holts      habtype  N agricultural
1       1       4     6      notpeat 47           no
2       3       2     0 agricultural 61          yes
3       4       1     8       cliffs 89           no
4       8       1     0       cliffs 89           no
5      11       1     0       cliffs 89           no
6      19       2     0 agricultural 61          yes
```

```
otterstrat <- svydesign(id = ~1, strata = ~habtype, fpc = ~N, data = otters)
svyby(~holts, by = ~agricultural, design = otterstrat, FUN = svytotal)
```

```
    agricultural    holts       se
```

```
no            no 877.9642 67.89958
yes          yes 106.7500 29.22259
```

This approach also works for simple random sampling.

```
notagg <- subset(ottersrs, habtype %in% c("cliffs","peat","notpeat"))
svytotal(~holts, design = notagg)
```

```
       total     SE
holts 1187.9 146.28
```

The estimate is different here because `svytotal` is using the estimator that does not use the (in this case known) size of the domain. But you can use the estimator that does know the size of the domain if you use post-stratification (as shown below).

Here is another approach using the `svycontrast` function which is quite general because it allows us to specify many different kinds of functions of stratum or domain means or totals. Here I will show how we can estimate the difference in the stratum means between the peat and non-peat strata.

```
tmp <- svyby(~holts, by = ~habtype, design = otterstrat, FUN = svymean)
tmp
```

```
                  habtype     holts        se
cliffs             cliffs  1.736842 0.4739725
agricultural agricultural  1.750000 0.4790589
peat                 peat 13.272727 1.0964954
notpeat           notpeat  4.095238 0.6408521
```

```
svycontrast(tmp, quote(peat - notpeat))
```

```
          nlcon   SE
contrast 9.1775 1.27
```

```
confint(svycontrast(tmp, quote(peat - notpeat)))
```

```
            2.5 %   97.5 %
contrast 6.688263 11.66672
```

Post-Stratification

Now assume that the sampling design was simple random sampling, but that we want to post-stratify based on the habitat auxiliary variable because we know how many sections are within each habitat. To do this we first need to create another data set that holds the known sizes of the strata.

```
habitatfreq <- data.frame(habtype = c("cliffs","agricultural","peat","notpeat"),
  Freq = c(89,61,40,47))
habitatfreq
```

```
       habtype Freq
1       cliffs   89
2 agricultural   61
3         peat   40
4      notpeat   47
```

This information can then be passed to the `postStratify` function which will effectively re-weight the observations.

```
otterpost <- postStratify(design = ottersrs, strata = ~habtype, population = habitatfreq)
```

Now we can estimate μ and τ as well as the domain means and totals.

```r
svymean(~holts, design = otterpost)
```

```
        mean     SE
holts 4.1549 0.3256
```

```r
svytotal(~holts, design = otterpost)
```

```
       total     SE
holts 984.71 77.162
```

```r
svyby(~holts, by = ~habtype, design = otterpost, FUN = svymean)
```

```
                  habtype     holts        se
cliffs             cliffs  1.736842 0.4232673
agricultural agricultural  1.750000 0.4634253
peat                 peat 13.272727 1.2994362
notpeat           notpeat  4.095238 0.6841975
```

```r
svyby(~holts, by = ~habtype, design = otterpost, FUN = svytotal)
```

```
                  habtype    holts       se
cliffs             cliffs 154.5789 37.67079
agricultural agricultural 106.7500 28.26895
peat                 peat 530.9091 51.97745
notpeat           notpeat 192.4762 32.15728
```

Note that the mean and total *estimates* are the same as those from stratified random sampling, but the standard errors are not. This is because the variances for estimators are not computed the same under post-stratification as they are under stratified random sampling. Note also that after post-stratification the estimator for a domain *total* uses the estimator that uses the known domain size. This also happens below where we estimate the number of holts in non-agricultural sections.

```r
notagg <- subset(otterpost, habtype %in% c("cliffs","peat","notpeat"))
svytotal(~holts, design = notagg)
```

```
       total     SE
holts 877.96 71.797
```

Note that the estimate is the same as when the design was specified as stratified random sampling, but the standard error is not.

Sampling Weights

We can compute sampling weights using the `weights` function. For simple random sampling we know that all elements have a weight of N/n.

```r
otters$w <- weights(ottersrs)
head(otters)
```

```
  section habitat holts      habtype  N agricultural        w
1       1       4     6      notpeat 47           no 2.890244
2       3       2     0 agricultural 61          yes 2.890244
3       4       1     8       cliffs 89           no 2.890244
4       8       1     0       cliffs 89           no 2.890244
5      11       1     0       cliffs 89           no 2.890244
6      19       2     0 agricultural 61          yes 2.890244
```

Remember that we changed the variable `N` earlier for the stratified random sampling design. For the weights for simple random sampling $N = 237$ and $n = 82$. For stratified random sampling an element has a weight of

N_j/n_j if it is form the j-th stratum.

```
otters$w <- weights(otterstrat)
head(otters)
```

```
  section habitat holts      habtype  N agricultural        w
1       1       4     6      notpeat 47           no 2.238095
2       3       2     0 agricultural 61          yes 3.050000
3       4       1     8       cliffs 89           no 4.684211
4       8       1     0       cliffs 89           no 4.684211
5      11       1     0       cliffs 89           no 4.684211
6      19       2     0 agricultural 61          yes 3.050000
```

When we use post-stratification we change the weights to match those for stratified random sampling.

```
otters$w <- weights(otterpost)
head(otters)
```

```
  section habitat holts      habtype  N agricultural        w
1       1       4     6      notpeat 47           no 2.238095
2       3       2     0 agricultural 61          yes 3.050000
3       4       1     8       cliffs 89           no 4.684211
4       8       1     0       cliffs 89           no 4.684211
5      11       1     0       cliffs 89           no 4.684211
6      19       2     0 agricultural 61          yes 3.050000
```

A property of sampling weights (if they have not been modified) is that they sum to the number of elements in the population (for simple random sampling) and to the number of elements in each stratum (for stratified random sampling).

```
otters <- otters %>%
  mutate(srswghts = weights(ottersrs), stratwghts = weights(otterstrat))
otters %>% summarize(weightsum = sum(srswghts))
```

```
  weightsum
1       237
```

```
otters %>% group_by(habtype) %>% summarize(weightsum = sum(stratwghts))
```

```
# A tibble: 4 x 2
  habtype      weightsum
  <fct>            <dbl>
1 cliffs              89
2 agricultural        61
3 peat                40
4 notpeat             47
```

There are useful ways that weights can be used that we will discuss later in the course.

Double Sampling (Two-Phase Sampling)

I am going to use another data set to demonstrate double sampling for strata with unknown sizes. The data are in a the **trtools** package which can be installed using the following command, provided you have already installed the **devtools** package (which can be installed using `install.packages(devtools)`).

```
devtools::install_github("trobinj/trtools")
```

The data set is called `ismail`. The elements are veterans. They were evaluated quickly in the first phase to classify them as disabled or not. Those veterans sampled in the second phase where then assessed by

psychiatrists to determine whether or not they had alcohol, sleep, or psychiatric disorders.

```
library(trtools)
ismail$N <- 53462
head(ismail, 20)
```

```
   disabled alcohol sleep psych     N
1        no    <NA>  <NA>  <NA> 53462
2        no    <NA>  <NA>  <NA> 53462
3        no    <NA>  <NA>  <NA> 53462
4        no    <NA>  <NA>  <NA> 53462
5        no    <NA>  <NA>  <NA> 53462
6        no    <NA>  <NA>  <NA> 53462
7        no    <NA>  <NA>  <NA> 53462
8        no    <NA>  <NA>  <NA> 53462
9        no    <NA>  <NA>  <NA> 53462
10       no    <NA>  <NA>  <NA> 53462
11      yes    <NA>  <NA>  <NA> 53462
12       no    <NA>  <NA>  <NA> 53462
13      yes      no   yes   yes 53462
14       no    <NA>  <NA>  <NA> 53462
15       no    <NA>  <NA>  <NA> 53462
16       no    <NA>  <NA>  <NA> 53462
17      yes      no    no    no 53462
18       no    <NA>  <NA>  <NA> 53462
19       no    <NA>  <NA>  <NA> 53462
20       no    <NA>  <NA>  <NA> 53462
```

The variable `disabled` is the stratification variable. The target variables (all categorical) are `alcohol`, `sleep`, and `psych`. The missing values (the `<NA>`) are due to the double sampling. First we specify a two-phase sampling design using `twophase`. Note that many of the arguments have two parts, one for each phase.

```
ismail2phase <- twophase(id = list(~1, ~1), strata = list(NULL, ~disabled),
  fpc = list(~N, NULL), subset = ~!is.na(sleep), data = ismail)
```

Inferences can then be obtained the usual way. Note that here because `sleep` is categorical, totals are the estimated *number* of veterans in the population with sleep disorders, and the means are the estimated *proportions* of veterans in the population with sleep disorders.

```
svytotal(~sleep, design = ismail2phase)
```

```
           total     SE
sleepno  44145.8 1826.1
sleepyes  9316.2 1826.1
```

```
svymean(~sleep, design = ismail2phase)
```

```
            mean     SE
sleepno  0.82574 0.0342
sleepyes 0.17426 0.0342
```